Network Science

Lecture 07: Small-World Networks

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

Speaker notes

This lecture asks two deceptively similar questions that turn out to be very different: (1) Why do short paths exist in large networks? (2) Can people *find* those short paths using only local information? The Watts-Strogatz model answers the first; Kleinberg's grid model answers the second. The gap between the two is the navigational gap — the fifth kind of gap in the course arc. Milgram's letter-chain experiment (1967) is the running example that threads the entire lecture.

Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

The letters that arrived took a **median of about 6 steps**.

The phrase "six degrees of separation" entered popular culture — but the experiment raised more questions than it answered.

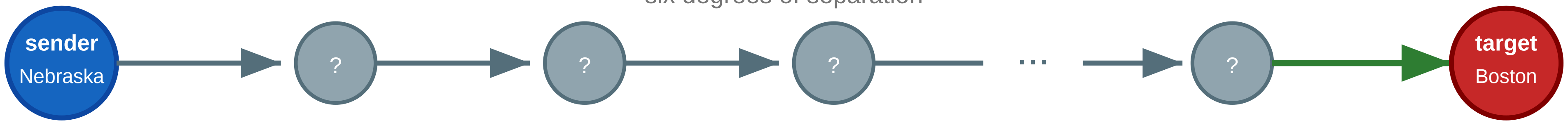
The Chain

Milgram's small-world experiment (1967)

Can a letter reach a stranger in Boston through a chain of personal acquaintances?

median ≈ 6 steps

"six degrees of separation"



Questions this experiment raises

① Why do short paths exist at all in a network of 200 million people?

② How can a network be locally clustered AND have short paths to strangers?

③ If short paths exist, can ordinary people find them using only local knowledge?

④ Only ~25% of chains arrived. Why did most fail?

Each intermediary sees only their own contacts and must guess who is "closer" to the target. About 25% of chains completed; the rest died when someone refused to forward. The experiment demonstrates both that short paths *exist* and that finding them is *hard*.

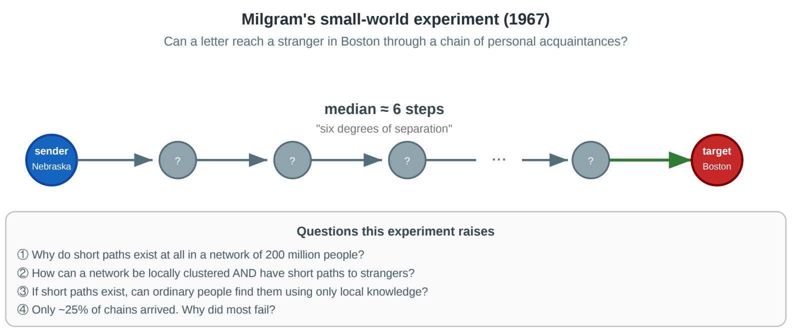
Speaker notes

This figure is the lecture anchor. Question 1 (why short paths?) leads to Module 1. Question 2 (clustering + short paths?) leads to Watts-Strogatz. Question 3 (can people find them?) leads to Kleinberg. Question 4 (why do most chains fail?) leads to the empirical anchor (Dodds et al. 2003). Refer back throughout.

Four Questions from One Experiment

1. **Existence.** Why do short paths exist at all in a network of 200 million people?
2. **Coexistence.** How can a network be *locally clustered* (your friends know each other) and *globally short* (six steps to a stranger in another state)?
3. **Navigability.** If short paths exist, can ordinary people find them using only local knowledge — no map, no search engine?
4. **Failure.** Only ~25% of chains arrived. Does this mean short paths are rare, or that *finding* them is the bottleneck?

Every concept in today's lecture answers one of these.



Speaker notes

Questions 1–2 map to Modules 1–2 (small-world phenomenon, Watts-Strogatz). Question 3 maps to Module 3 (Kleinberg). Question 4 maps to Module 4 (Dodds et al. empirical test). The Web module (Module 5) extends the navigability question to directed information networks.

A New Kind of Gap — Existence vs. Findability

L03–L04 had a **computational** gap (NP-hard ideals). L05 had a **causal** gap. L06 had a **structural** gap (completeness required).

L07 has a **navigational** gap: short paths may *exist* without anyone being able to *find* them using local information.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance
L07	Navigational	Short navigable paths	Local information only	Geometric link distribution

Speaker notes

The navigational gap is structurally different from the previous ones: it is not about what we can *compute* or *measure*, but about what *actors inside the network* can do with limited information. The fix is not a better algorithm run by an external analyst — it is a specific *geometry* of long-range links that makes local decisions globally effective. Kleinberg's result identifies this geometry precisely.

Takeaway

The small-world phenomenon is really two phenomena: the *existence* of short paths (explained by a few random long-range links) and the *findability* of those paths (explained only by a specific geometric distribution of links). Milgram's experiment revealed both — and the gap between them.

The Small-World Phenomenon

> **Guiding Question:** Why are distances in large networks often surprisingly short?

Speaker notes

This module establishes the basic empirical fact — typical distances grow logarithmically with network size — and gives examples from social, collaboration, and communication networks.

Logarithmic Distances

Small-world property. A network exhibits the small-world property if the average shortest-path distance $\bar{d}$ grows at most logarithmically with the number of nodes:

$$\bar{d} \propto \log |V|$$

In a random graph $G(n, p)$ with average degree k, the typical distance is approximately $\bar{d} \approx \frac{\log n}{\log k}$.

Speaker notes

The logarithmic scaling is the formal content of "six degrees of separation." It explains why Milgram's result is not mysterious from a graph-theory perspective: in any network where each node has tens or hundreds of contacts, the branching factor is so high that $\log n / \log k$ remains small even for billions of nodes. The surprise is not that short paths exist — it is that ordinary people can sometimes find them.

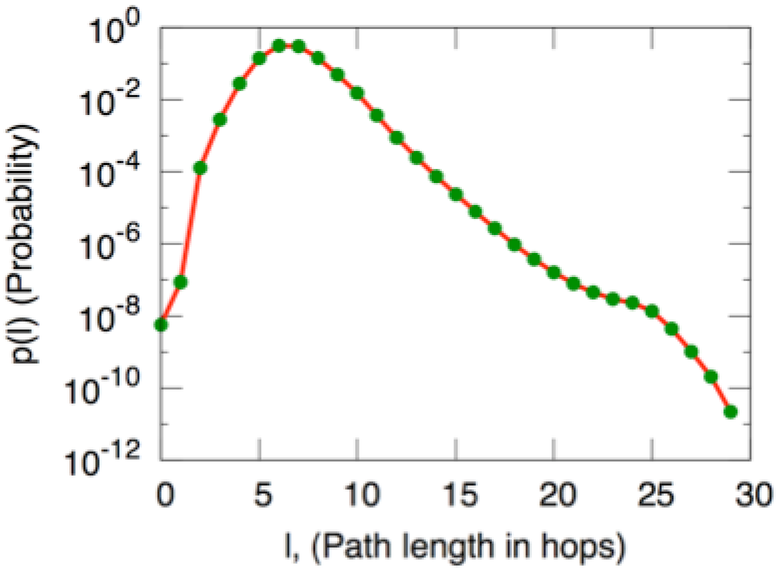
Logarithmic Distances

> **Small-world property.** A network exhibits the small-world property if the average shortest-path distance $\bar{d}$ grows at most logarithmically with the number of nodes:
>
> $$\bar{d} \propto \log |V|$$
>
> In a random graph $G(n, p)$ with average degree k, the typical distance is approximately $\bar{d} \approx \frac{\log n}{\log k}$.

For a social network of $n = 300$ million people with average degree $k \approx 100$: $\bar{d} \approx \frac{\log(3 \times 10^8)}{\log 100} \approx \frac{19.5}{4.6} \approx 4.2$. Milgram's 6 is in the right ballpark.

≡

Empirical Examples



Source: Easley & Kleinberg 2010, p. Ch. 20

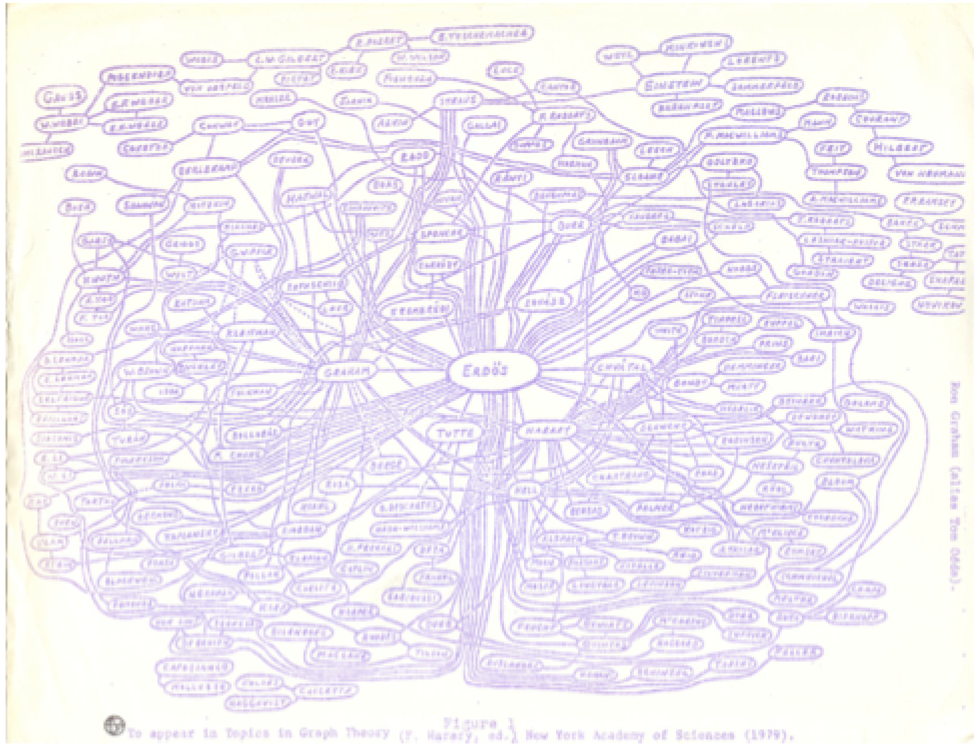


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdös [189]. (Image from http://www.oakland.edu/enp/cgraph.jpg)

Source: Easley & Kleinberg 2010

Left: Microsoft Instant Messenger network — 180 million users, median distance ~7. **Right:** Mathematical collaboration network — Erdős numbers rarely exceed 5. Both confirm logarithmic scaling across orders of magnitude.

Speaker notes

The IM data (Leskovec & Horvitz 2008) is especially striking: 180 million nodes, 1.3 billion edges, average distance 6.6. The Erdős collaboration graph is smaller (~400k mathematicians) but culturally iconic. Both examples reinforce that logarithmic distances are a robust empirical fact, not a quirk of Milgram's Nebraska sample.

Takeaway

> Logarithmic distances are a robust empirical fact across social, communication, and collaboration networks. In any network with moderate average degree, even billions of nodes remain only a handful of hops apart.

Quick Check

The Facebook social graph has approximately 3 billion users and an average degree of about 300.

1. Estimate the average shortest-path distance using $\bar{d} \approx \log n / \log k$.
2. Facebook reported an empirical average distance of 3.57 in 2016. Is this consistent with your estimate?
3. Why might the empirical value be *lower* than the logarithmic estimate?

Speaker notes

Give them 3 minutes. (1) $\bar{d} \approx \log(3 \times 10^9)/\log(300) \approx 21.8/5.7 \approx 3.8$. (2) Yes — 3.57 is close to 3.8. (3) The formula assumes a random graph; Facebook has hubs (celebrities, organizations) with degrees far above 300, which create even shorter paths. The degree distribution is heavy-tailed, not Poisson, so a few high-degree nodes pull the average distance below the random-graph prediction.

Quick Check — Solution

$$\bar{d} \approx \frac{\log(3\times10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx \mathbf{3.8}$$

Yes — 3.57 is remarkably close to the logarithmic estimate.

The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

Quick Check — Solution

1. $\bar{d} \approx \frac{\log(3\times10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx \mathbf{3.8}$

Yes — 3.57 is remarkably close to the logarithmic estimate.

The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

Quick Check — Solution

1. $\bar{d} \approx \frac{\log(3\times10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx 3.8$

2. Yes — 3.57 is remarkably close to the logarithmic estimate.

The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

Quick Check — Solution

1. $\bar{d} \approx \frac{\log(3\times10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx 3.8$

2. Yes — 3.57 is remarkably close to the logarithmic estimate.

3. The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

The Watts–Strogatz Model

> **Guiding Question:** How can a network be both highly clustered *and* have short paths?

Speaker notes

This is the paradox that Watts & Strogatz (1998) resolved. Regular lattices have high clustering but long paths. Random graphs have short paths but no clustering. Real social networks have *both*. The Watts-Strogatz model shows that a tiny amount of randomness bridges the gap.

The Paradox

Real social networks have two seemingly contradictory properties:

Property	Regular lattice	Random graph	Real social network
Clustering C	High (~ 0.5)	Low ($\sim k/n$)	**High** (~ 0.1–0.5)
Avg. path length L	Long ($\sim n/2k$)	Short ($\sim \log n / \log k$)	**Short** ($\sim \log n$)

Speaker notes

The table makes the paradox concrete. The challenge is to find a model that produces the bottom-right cell: high clustering AND short paths. Neither a lattice nor a random graph does this alone. The Watts-Strogatz model interpolates between the two extremes with a single parameter p.

The Paradox

Real social networks have two seemingly contradictory properties:

Property	Regular lattice	Random graph	Real social network
Clustering C	High (~ 0.5)	Low ($\sim k/n$)	**High** (~ 0.1–0.5)
Avg. path length L	Long ($\sim n/2k$)	Short ($\sim \log n/\log k$)	**Short** ($\sim \log n$)

A regular lattice is clustered but large. A random graph is small but unclustered. **Real networks are both** — high C and low L simultaneously.

The Model

Watts–Strogatz model (1998).

1. Start with a **ring lattice**: n nodes arranged in a circle, each connected to its k nearest neighbors.
2. For each edge, with probability p, **rewire** one endpoint to a uniformly random node (avoiding self-loops and duplicates).

The parameter $p \in [0, 1]$ controls the interpolation:

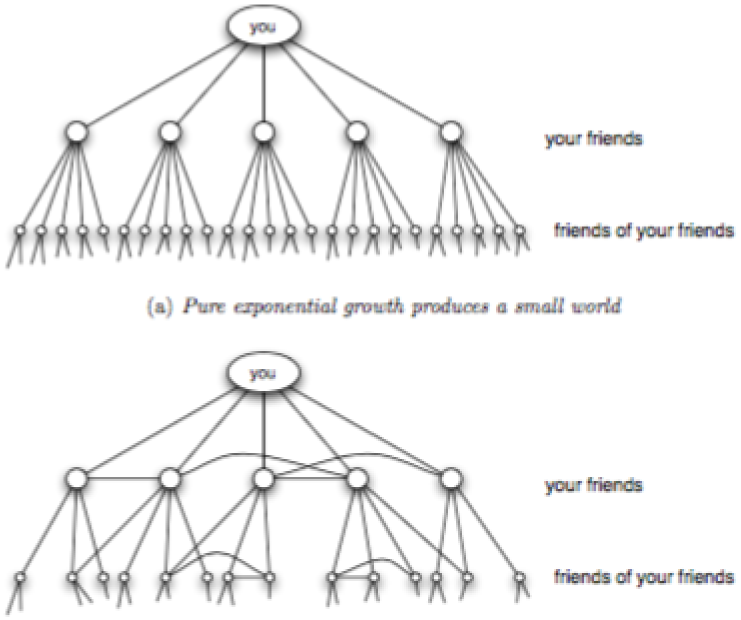
- $p = 0$: regular lattice (high C, high L)
- $p = 1$: random graph (low C, low L)
- $0 < p \ll 1$: **small-world regime** (high C, low L)

≡

Speaker notes

The model is beautifully simple: one parameter, one structural transition. The key insight is that the transition in $L(p)$ and $C(p)$ happens at *different* speeds. Path length collapses rapidly (a few random shortcuts suffice) while clustering decays slowly (most triangles survive). This creates a wide window where $C(p) \approx C(0)$ but $L(p) \approx L(1)$ — the small-world regime.

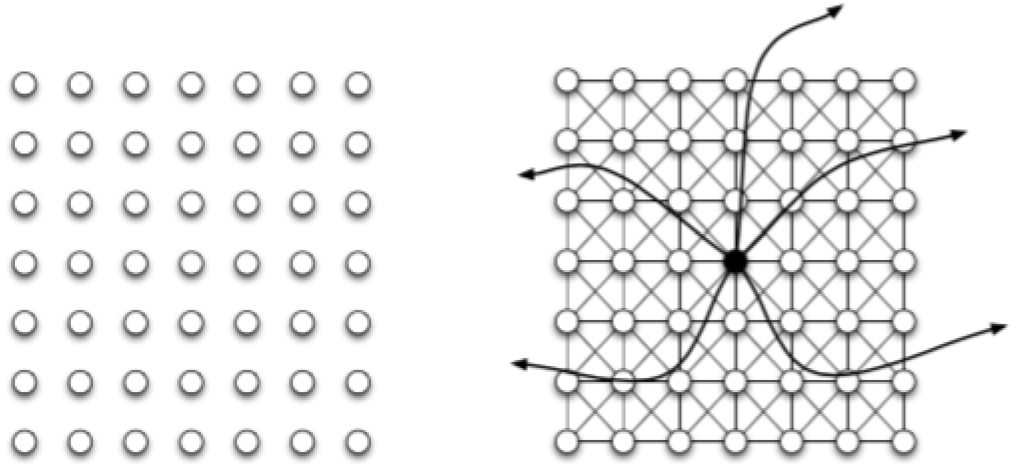
The Characteristic Curves



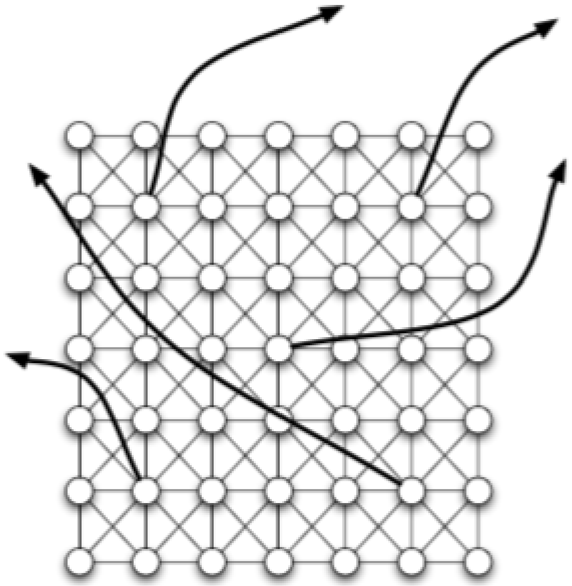
(a) *Pure exponential growth produces a small world*

(b) *Triadic closure reduces the growth rate*

Source: Easley & Kleinberg 2010, p. Ch. 20



Source: Easley & Kleinberg 2010, p. Ch. 20



Source: Easley & Kleinberg 2010, p. Ch. 20

$C(p)/C(0)$ stays near 1 while $L(p)/L(0)$ drops sharply. The small-world regime (shaded) spans roughly $0.001 \leq p \leq 0.1$ – a few percent of edges rewired is enough to make the world small.

Speaker notes

The $C(p)/L(p)$ plot is the single most important figure of the module. The visual punchline: the L curve drops like a cliff while the C curve barely moves. This asymmetry is what makes the small-world regime possible. In practical terms: rewiring just 1% of edges in a lattice reduces average path length to near-random-graph levels while preserving almost all clustering.

Takeaway

The Watts-Strogatz model resolves the clustering/path-length paradox: a tiny fraction of random long-range shortcuts collapses distances without destroying local structure. The small-world regime exists because $L(p)$ falls much faster than $C(p)$.

Quick Check

A ring lattice has $n = 1000$ nodes and $k = 10$ neighbors per node.

1. Estimate the average path length L at $p = 0$.
2. After rewiring 1% of edges ($p = 0.01$), how many long-range shortcuts are created?
3. Qualitatively, what happens to L and C?

Speaker notes

Give them 3 minutes. (1) $L(0) \approx n/(2k) = 1000/20 = 50$. (2) The lattice has $nk/2 = 5000$ edges; 1% rewired = 50 long-range shortcuts. (3) L drops dramatically — 50 random shortcuts in a 1000-node graph is enough to bring L close to $\log(1000)/\log(10) \approx 3$. C drops only slightly — most of the $\binom{k}{2}$ triangles per node survive because only 50 out of 5000 edges moved.

Quick Check — Solution

$L(0) \approx n/(2k) = 1000/20 = 50$ steps.

2. $nk/2 = 5000$ edges total; 1% =   .

3. L collapses — 50 random shortcuts in a 1000-node lattice bring L close to $\log n/\log k \approx 3$. C barely changes — only 50 of 5000 edges were rewired, so the vast majority of local triangles survive.

Quick Check — Solution

1. $L(0) \approx n/(2k) = 1000/20 = 50$ steps.

2. $nk/2 = 5000$ edges total; 1% = .

3. L collapses — 50 random shortcuts in a 1000-node lattice bring L close to $\log n / \log k \approx 3$. C barely changes — only 50 of 5000 edges were rewired, so the vast majority of local triangles survive.

Quick Check — Solution

1. $L(0) \approx n/(2k) = 1000/20 = 50$ steps.

2. $nk/2 = 5000$ edges total; 1% = **50 long-range shortcuts**.

3. L collapses — 50 random shortcuts in a 1000-node lattice bring L close to $\log n/\log k \approx 3$. C barely changes — only 50 of 5000 edges were rewired, so the vast majority of local triangles survive.

Quick Check — Solution

1. $L(0) \approx n/(2k) = 1000/20 = 50$ steps.

2. $nk/2 = 5000$ edges total; 1% = **50 long-range shortcuts**.

3. L collapses — 50 random shortcuts in a 1000-node lattice bring L close to $\log n/\log k \approx 3$. C barely changes — only 50 of 5000 edges were rewired, so the vast majority of local triangles survive. **This is the small-world regime.**

Kleinberg's Decentralized Search

> **Guiding Question:** Short paths exist — but can local actors *find* them without a map?

Speaker notes

This is where the navigational gap opens. Watts-Strogatz proved that short paths exist; Kleinberg (2000) asked whether a *decentralized* agent — one that sees only its own neighbors and knows the target's approximate location — can find them efficiently. The answer is: only if the long-range links have a very specific geometric distribution.

The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

Speaker notes

This is the conceptual crux of the lecture. Existence of short paths is a property of the graph. Findability of short paths is a property of the *algorithm plus the graph*. Watts-Strogatz gives existence; Kleinberg gives the conditions for findability.

The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

Decentralized search: at each step, the current holder forwards to the neighbor *closest* to the target (in some distance metric). No global routing table, no BFS.

The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

Decentralized search: at each step, the current holder forwards to the neighbor *closest* to the target (in some distance metric). No global routing table, no BFS.

Watts-Strogatz does not guarantee this works. Short paths exist, but the greedy algorithm may not find them — it depends on *where* the long-range links point.

≡

Kleinberg's Grid Model (2000)

Setup. Place n nodes on a d-dimensional grid. Each node has:

- **local contacts:** all grid neighbors within lattice distance p
- **one long-range link:** drawn with probability proportional to $d(v, u)^{-r}$, where d is grid distance and $r \geq 0$ is the distance exponent.

Greedy routing. At each step, forward to the neighbor (local or long-range) closest to the target in grid distance.

The parameter r controls the *geometry* of long-range links:

- $r = 0$: uniform random (long-range links go anywhere)
- $r = d$: inverse-power law matched to dimension (links span all scales equally)
- $r \gg d$: long-range links are effectively local

≡

Speaker notes

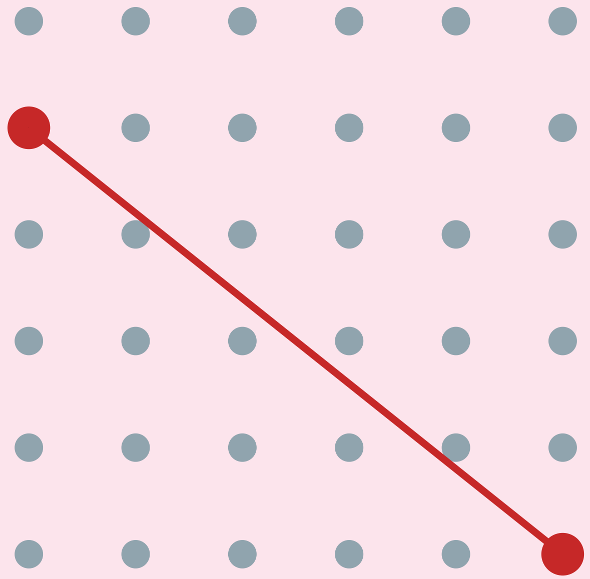
The model separates the two ingredients cleanly: the grid provides the local clustering, and the long-range links provide the shortcuts. The exponent r is the single knob that controls navigability. Kleinberg's result is that only one value of r makes greedy search efficient.

The Three Regimes

Kleinberg's grid model — the geometry of navigability

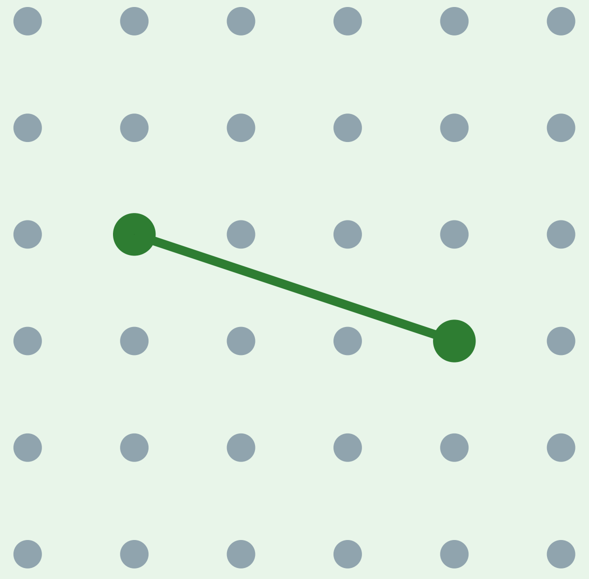
Each node has local grid neighbors + one long-range link drawn with $P(\text{link to } u) \propto d(v,u)^{-r}$

$r = 0$ (uniform random)



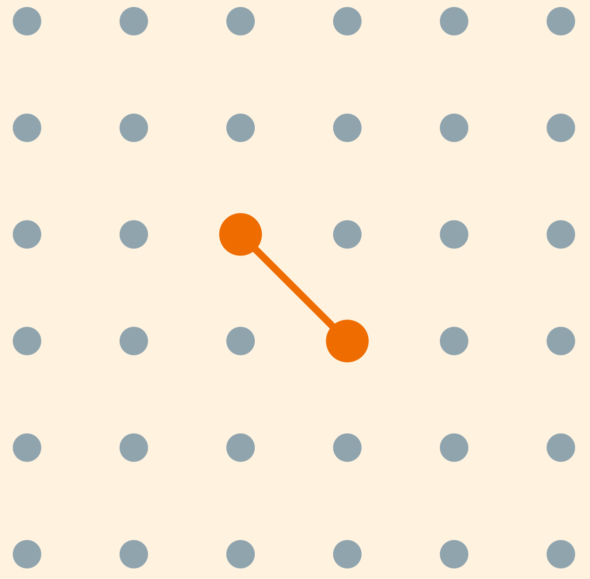
short paths exist,
but search is $\Omega(n^{2/3})$

$r = 2 = d$ (inverse-square)



short paths exist
AND search is $O(\log^2 n)$

$r \gg d$ (too clustered)



short paths exist,
but "shortcuts" are too local

Speaker notes

The figure shows why $r = d$ is special. At $r = 0$, long-range links are uniformly random — they create short paths but provide no geographic gradient for the greedy algorithm to follow. At $r \gg d$, links are so local that they add no useful shortcuts. Only at $r = d$ do the links span all scales proportionally, creating a hierarchy of shortcuts that the greedy algorithm can exploit at every distance.

Kleinberg's Theorem

Theorem (Kleinberg, 2000).

In the d-dimensional grid model with long-range exponent r:

- If $r = d$: greedy routing finds the target in $O(\log^2 n)$ steps.
- If $r \neq d$: **no** decentralized algorithm (using only local information) can find the target in fewer than $\Omega(n^c)$ steps for some constant $c > 0$.

≡

Speaker notes

This is one of the sharpest results in the course. The navigability transition is not gradual — it is a threshold phenomenon. At $r = d$, greedy search succeeds in $O(\log^2 n)$ steps; at any other r, it is polynomial in n. The uniqueness of $r = d$ means that navigability is not a generic property of small-world networks — it requires a specific geometry that most random-shortcut models (including Watts-Strogatz) do not guarantee.

Kleinberg's Theorem

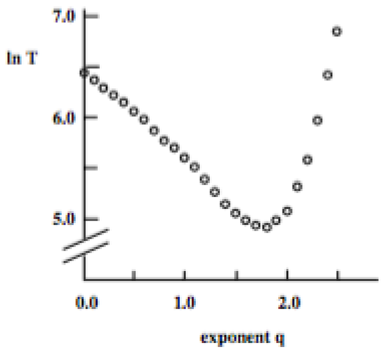
Theorem (Kleinberg, 2000).

In the d-dimensional grid model with long-range exponent r:

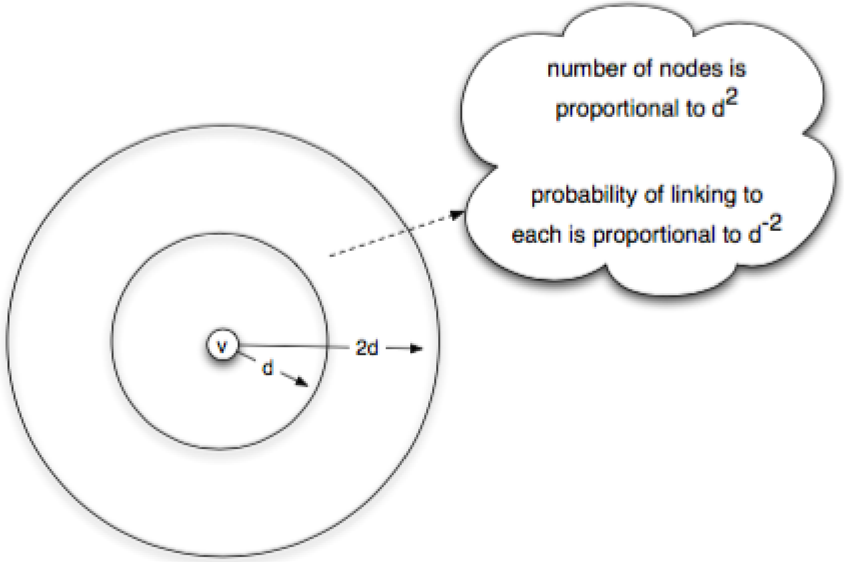
- If $r = d$: greedy routing finds the target in $O(\log^2 n)$ steps.
- If $r \neq d$: **no** decentralized algorithm (using only local information) can find the target in fewer than $\Omega(n^c)$ steps for some constant $c > 0$.

The exponent $r = d$ is **unique** — it is the only value at which decentralized search is polylogarithmic. Any other value, and search degrades to polynomial.

Simulation Evidence



Source: Easley & Kleinberg 2010, p. Ch. 20



Source: Easley & Kleinberg 2010, p. Ch. 20

Left: delivery time vs. exponent r — a sharp minimum at $r = d = 2$. **Right:** the proof intuition — at $r = d$, each "ring" of nodes at distance 2^i from the target receives roughly equal link probability, creating a hierarchy of scales the algorithm descends one level at a time.

Speaker notes

The simulation plot is the visual proof that the theorem is not just asymptotic: even on moderate-sized grids (hundreds to thousands of nodes), the minimum at $r = d$ is sharp and clearly visible. The "equal probability per ring" intuition is the heart of the proof: if each doubling of distance carries equal total link probability, then a greedy step reduces the remaining distance by a constant factor with constant probability, yielding $O(\log n)$ rounds of $O(\log n)$ steps each.

Takeaway

> Short paths exist in almost any small-world network. But navigability — the ability of local actors to *find* those paths — requires a specific geometric distribution of long-range links ($r = d$). Watts-Strogatz gives existence; Kleinberg gives findability.

Quick Check

A social network is well-modeled by a 2D grid with random long-range links.

1. What exponent r makes greedy search efficient?
2. If the network's long-range links are uniformly random ($r = 0$), what happens to (a) the existence of short paths and (b) greedy routing performance?
3. Why is the Watts-Strogatz model insufficient to explain Milgram's result?

≡

Speaker notes

Give them 4 minutes. (1) $r = 2$ (matches the grid dimension $d = 2$). (2a) Short paths still exist — random shortcuts reduce L. (2b) Greedy routing degrades to polynomial time — the shortcuts point to random locations that provide no useful gradient toward the target. (3) Watts-Strogatz shows short paths *exist* but uses uniform rewiring ($r = 0$), which does not support efficient decentralized search. Milgram's participants actually *found* short paths, which requires $r = d$ — a stronger condition than Watts-Strogatz guarantees.

Quick Check — Solution

$r = 2 = d$ for a 2D grid.

- (a) Short paths — uniform random shortcuts reduce average path length.
- (b) Greedy routing — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

- (a) Short paths — uniform random shortcuts reduce average path length.
- (b) Greedy routing — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

≡

Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths — uniform random shortcuts reduce average path length.
- (b) Greedy routing — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths **still exist** — uniform random shortcuts reduce average path length.
- (b) Greedy routing — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths **still exist** — uniform random shortcuts reduce average path length.
- (b) Greedy routing **fails** — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths **still exist** — uniform random shortcuts reduce average path length.
- (b) Greedy routing **fails** — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not *navigable* short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

≡

Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

Speaker notes

Bridge slide. The students have just proved that greedy routing works at $r = d$. The same mathematical structure — hierarchy of scales, greedy descent — reappears in a completely different setting: approximate nearest-neighbour (ANN) search in high-dimensional vector spaces. This is a running theme of the course: a structural principle discovered in one domain turns out to be load-bearing in another.

Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

The grid gave us a **known, low-dimensional** metric. What happens when the "distance" is something we have to *learn* from data — high-dimensional embeddings of images, documents, or molecules?

Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

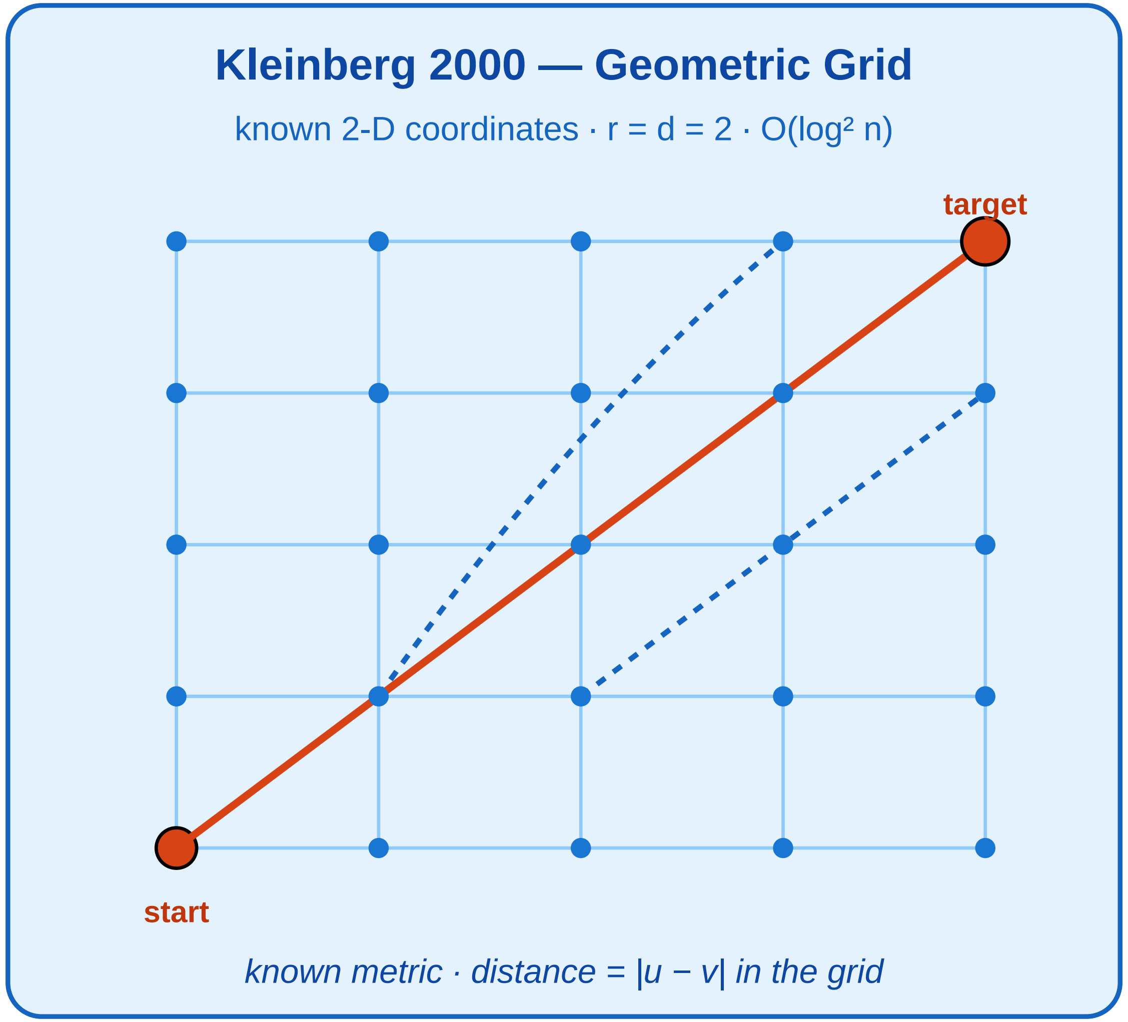
The grid gave us a **known, low-dimensional** metric. What happens when the "distance" is something we have to *learn* from data — high-dimensional embeddings of images, documents, or molecules?

> The same small-world recipe powers **modern vector search**: the index graphs used inside Pinecone, Weaviate, Milvus, FAISS, and every retrieval-augmented LLM pipeline are Kleinberg's grid in disguise.

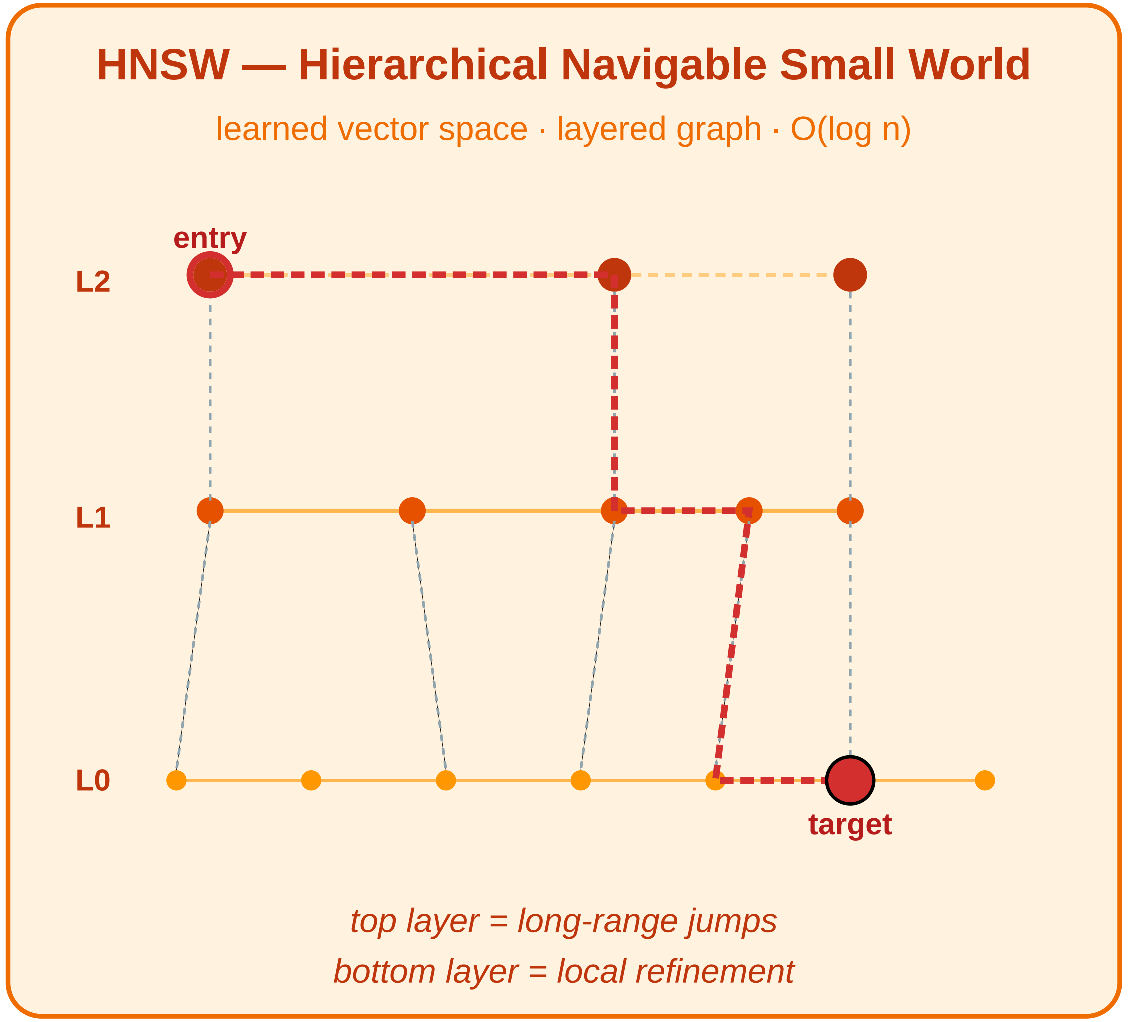
From Kleinberg's Grid to HNSW

From Kleinberg's Grid to Modern Vector Search (HNSW)

Same principle: long-range + short-range links enable greedy routing



generalise metric → high-dim vectors



Left: Kleinberg's 2-D grid — known coordinates, greedy routing with $r = d = 2, O(\log^2 n)$ delivery. **Right:** Hierarchical Navigable Small World (HNSW) — learned vectors, multi-layer graph, greedy descent from sparse top layer to dense bottom layer, $O(\log n)$ query time.

Speaker notes

HNSW (Malkov & Yashunin 2016/2018) is the single most widely deployed ANN algorithm today. Its architecture is literally Kleinberg's insight turned into an engineering artefact: the top layer contains a few nodes connected by long-range jumps, each lower layer adds shorter edges, and greedy search descends the hierarchy. The "navigability" proof in Kleinberg becomes an engineering guarantee in HNSW. The mapping is: coordinate grid → learned embedding space; $r = d$ exponent → explicit multi-scale layering; local/long-range split → layered edge structure.

HNSW in Three Ideas

1. Layers as scales. Each node is assigned a maximum level $l \sim \mathrm{Geom}(1/\ln M)$. Layer k contains only nodes with level $\geq k$ — so higher layers are exponentially sparser.

2. Greedy per layer. Search starts at the top-layer entry point, greedily moves to the neighbour closest to the query, then descends one layer and repeats — refining locally.

3. Long-range by construction. Top-layer edges span the whole space (like Kleinberg's $r = d$ links); bottom-layer edges are local. The geometric distribution is *baked into the index*, not drawn from a random process.

Speaker notes

The geometric level assignment is the key design choice: it gives the index the same multi-scale distribution of links that Kleinberg proved optimal, without requiring knowledge of the "dimension" of the embedding space. Teachers sometimes introduce HNSW as if it were a pure data-structures result; in fact it is a direct algorithmic descendant of Kleinberg 2000. The exponent $r = d$ and the level-assignment geometric distribution play the same role: guaranteeing that each "doubling" of distance has roughly equal link probability.

HNSW in Three Ideas

1. Layers as scales. Each node is assigned a maximum level $l \sim \mathrm{Geom}(1/\ln M)$. Layer k contains only nodes with level $\geq k$ — so higher layers are exponentially sparser.

2. Greedy per layer. Search starts at the top-layer entry point, greedily moves to the neighbour closest to the query, then descends one layer and repeats — refining locally.

3. Long-range by construction. Top-layer edges span the whole space (like Kleinberg's $r = d$ links); bottom-layer edges are local. The geometric distribution is *baked into the index*, not drawn from a random process.

Complexity. $O(\log n)$ query time in practice; billions of vectors routinely indexed in production.

≡

Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into $\mathbb{R}^d$ (typically $d = 384$–4096).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
3. At query time, **greedy-search** from the entry point for the top-k nearest neighbours.

Speaker notes

This is the most important forward-looking slide in the lecture. Students should leave understanding that the theoretical result from 2000 is load-bearing for the infrastructure they use daily. The "navigational gap" from L07 becomes a *design principle* for index structures. Forward-reference L09, which introduces the embedding side of the story — how the vectors that live in this index are produced in the first place. The complete picture (embedding + index) requires both lectures.

Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into $\mathbb{R}^d$ (typically $d = 384–4096$).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
3. At query time, **greedy-search** from the entry point for the top-k nearest neighbours.

A 2026-era LLM stack searching 100M+ embeddings in <10 ms is doing *Kleinberg-style decentralized search* — on a learned metric, in a graph someone built to guarantee the $r = d$ property.

Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into $\mathbb{R}^d$ (typically $d = 384–4096$).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
3. At query time, **greedy-search** from the entry point for the top-k nearest neighbours.

A 2026-era LLM stack searching 100M+ embeddings in <10 ms is doing *Kleinberg-style decentralized search* — on a learned metric, in a graph someone built to guarantee the $r = d$ property.

> We will come back to **how** these embeddings are constructed in **Lecture 09** (node and document representations, spectral methods, DeepWalk, node2vec, GNNs). The *geometry* of the embedding space determines whether navigation works — which closes the loop to Kleinberg.

Empirical Anchor: The Global Email Experiment

> **Guiding Question:** Does Milgram's result replicate at scale — and what does chain failure tell us?

Speaker notes

This module provides the empirical anchor for L07, following the pattern of Kossinets-Watts (L03), Zachary (L04), Christakis-Fowler (L05), and Leskovec et al. (L06). The study is Dodds, Muhamad & Watts (2003), the largest replication of Milgram's experiment.

The Study

Dodds, Muhamad & Watts (2003), *An Experimental Study of Search in Global Social Networks*, [Science 301, 827–829](#).

They replicated Milgram's experiment using **email** — a global medium with no geographic constraint.

Parameter	Value
Participants recruited	**~60 000** from 166 countries
Target persons	**18** in 13 countries
Chains started	**24 163**
Chains completed	**384** (~1.6%)
Median chain length (completed)	**4–7 steps**

≡

Speaker notes

The scale is two orders of magnitude larger than Milgram's 296 letters, and the medium (email) removes geographic friction. The completed chains confirm logarithmic distances (~5 steps median). But the headline number is the *completion rate*: only 1.6% of chains reached their target. This is even lower than Milgram's ~25%, despite the ease of forwarding email.

What They Found

1. Completed chains are short. Median 4–7 steps, consistent with Milgram and with $\log n / \log k$ scaling.

2. Most chains die early. The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.

Speaker notes

Finding 3 is the empirical bridge to Kleinberg: successful Milgram-style search is not random — it exploits social coordinates (geography, profession, organizational hierarchy) that function as approximate distance metrics. The chains that succeeded were the ones where each forwarder made a good *greedy* decision relative to the target's known attributes. This is exactly what Kleinberg's theorem predicts: navigability requires that long-range links are distributed across scales in a way that local greedy decisions can exploit.

What They Found

1. Completed chains are short. Median 4–7 steps, consistent with Milgram and with $\log n / \log k$ scaling.

2. Most chains die early. The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.

3. Successful chains use geography and profession as search dimensions. Forwarders who brought the letter closer in *geographic* or *professional* distance were more effective — consistent with Kleinberg's model where search exploits multiple distance dimensions simultaneously.

What the Study Could Not See

1. Chain death ≠ path absence. A chain that dies after 3 steps does not mean no short path existed — it means someone chose not to forward. The study measures *willingness to search*, not *path existence*. The true completion rate under full cooperation is unknown.

2. Selection bias in completions. The 384 completed chains are a biased sample — they may systematically over-represent easy targets (well-known, geographically close, professionally distinctive). The median chain length of 4–7 steps may underestimate the true difficulty of reaching a random target.

3. Email ≠ acquaintance. Milgram's experiment required *personal acquaintance*; the email study allowed forwarding to anyone whose email address you knew. The social network traversed by email may be larger and weaker-tied than the face-to-face network Milgram assumed.

4. No ground-truth network. Without observing the full network, the study cannot measure whether the successful chains actually followed near-optimal paths or just happened to complete.

Speaker notes

The chain-death problem is the deepest limitation: the experiment conflates the *ability* to find short paths with the *willingness* to search. Milgram's original experiment had the same issue. No routing experiment can fully separate topology from motivation without observing the complete network and making forwarding mandatory — which is ethically impossible. This is the navigational gap in its purest empirical form.

Takeaway

Dodds et al. (2003) confirmed Milgram at scale: short paths exist and people can sometimes find them. But the 1.6% completion rate reveals that *motivation*, not topology, is the primary bottleneck. The navigational gap is real — short paths exist but are rarely found in practice.

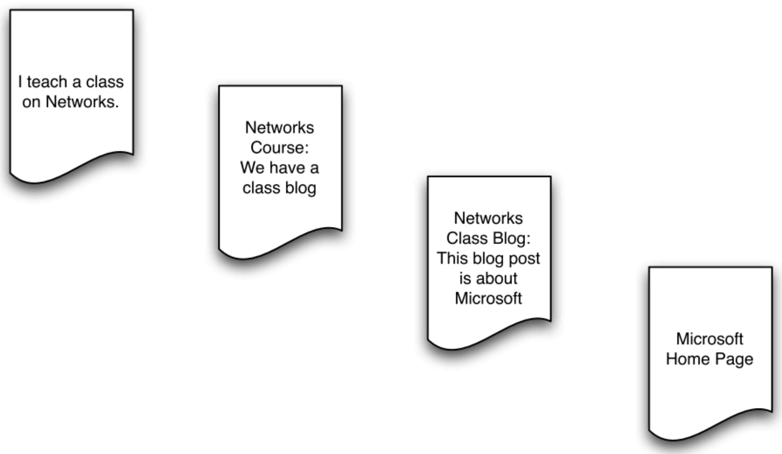
The Web as a Directed Small World

Guiding Question: What happens to navigability when links are directed?

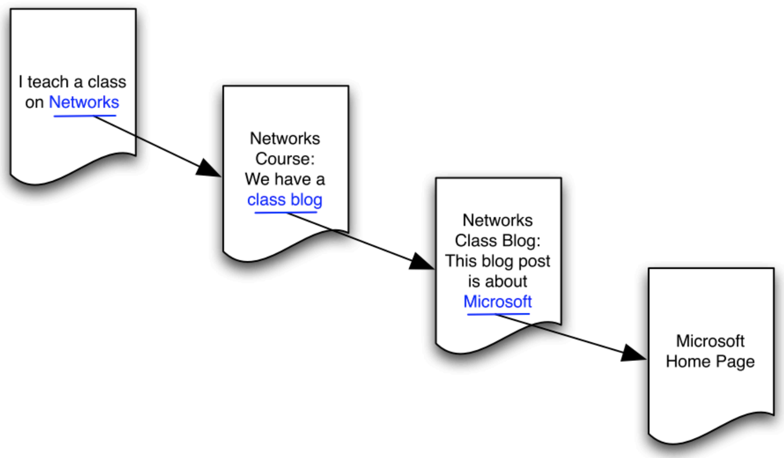
Speaker notes

The Web is the largest information network ever built, and its links are directed — a page can link to another without being linked back. This breaks the symmetry that underpins both Watts-Strogatz and Kleinberg: in a directed graph, reachability is not mutual, and the "small world" may only exist in part of the network.

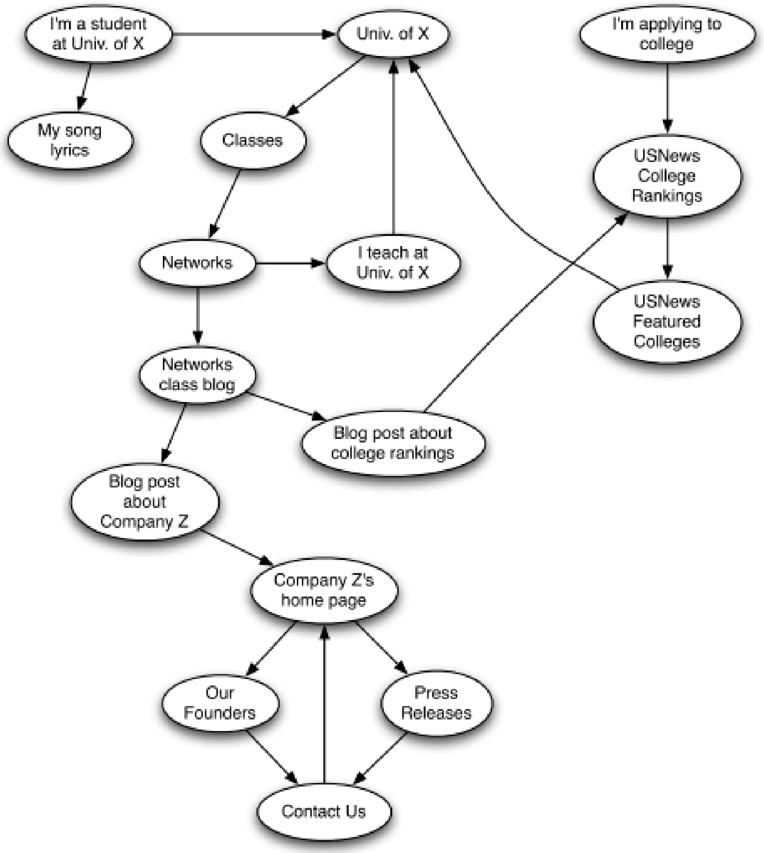
From Social to Information Networks



Source: Easley & Kleinberg 2010



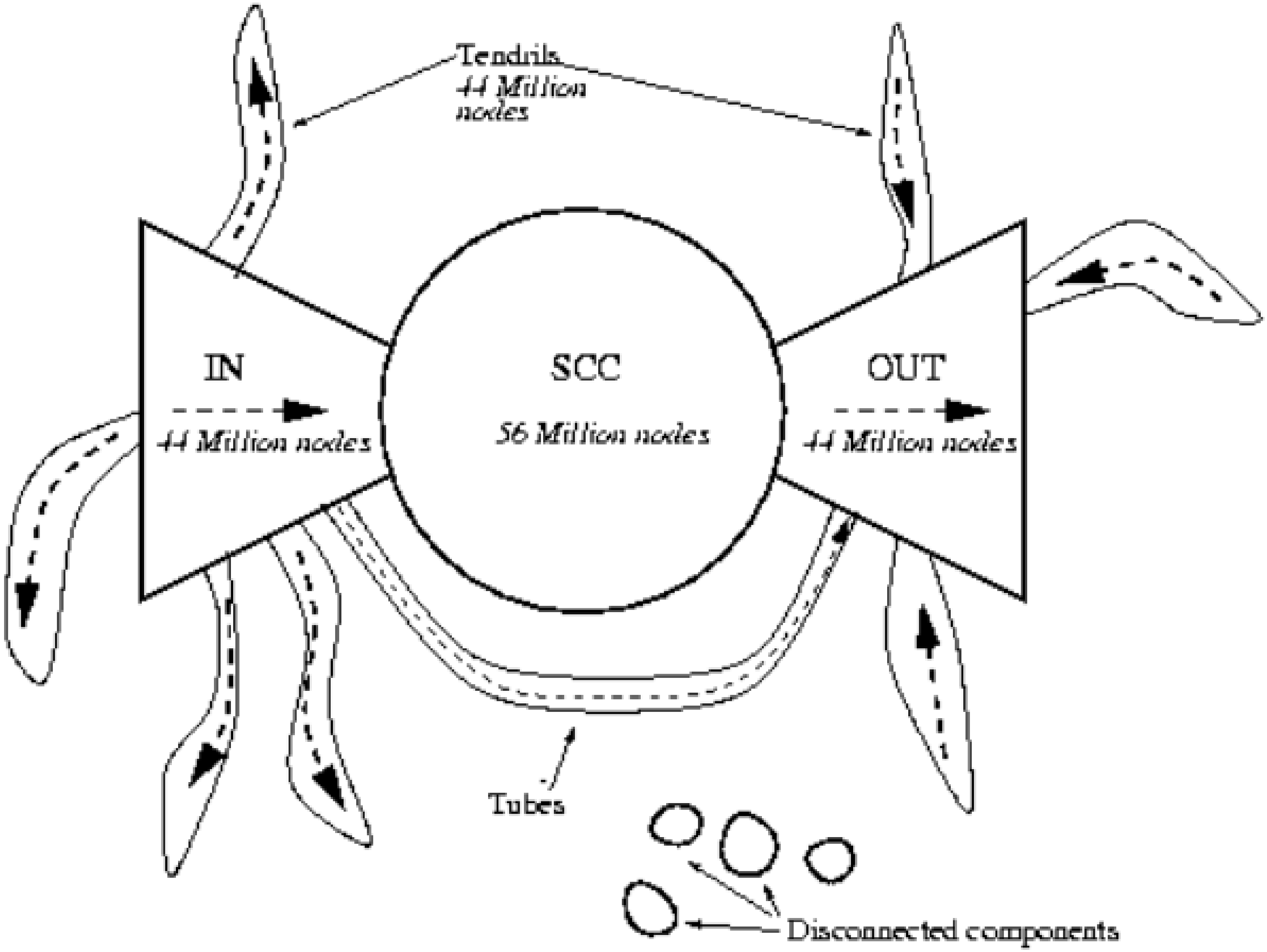
Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

Hyperlinks are **directed** — page A can link to page B without B linking back. This means reachability is asymmetric: you can follow links *from* A *to* B but not necessarily in reverse.

The Bow-Tie Structure



Source: Easley & Kleinberg 2010, p. Ch. 13

Broder et al. (2000) crawled ~200 million web pages and found the Web decomposes into:

- **SCC (Strongly Connected Component):** ~28% of pages — every page reachable from every other via directed paths. *This is the "small world" of the Web.*
- **IN:** pages that can reach the SCC but are not reachable from it.
- **OUT:** pages reachable from the SCC but that cannot reach back.
- **Tendrils and tubes:** pages connected to IN or OUT but not to the SCC.

Speaker notes

The bow-tie decomposition is the directed analogue of connected components. The SCC is the only part where mutual reachability holds — where "six degrees" makes sense. The IN, OUT, and tendril components break the small-world property because paths exist in one direction but not the other. Roughly 72% of the Web is *not* in the SCC, meaning the "small world" is a minority of the full graph.

Navigability in Directed Networks

The small-world property and navigability both change in directed networks:

- **Short paths exist** within the SCC — the strongly connected core is a small world (~16 clicks between random page pairs within SCC).
- **Short paths do not exist** from OUT back to SCC, or between tendrils.
- **Greedy search is harder** because you can only follow outgoing links — you cannot "pull" information from a page that doesn't link to you.

Speaker notes

This connects to L04: PageRank (eigenvector centrality on the directed Web graph) is the tool that makes centralized Web search possible. The Web is simultaneously a small world (within the SCC) and not navigable by local greedy search (because of directionality). The resolution — a centralized search engine — sidesteps the navigational gap rather than solving it.

Navigability in Directed Networks

The small-world property and navigability both change in directed networks:

- **Short paths exist** within the SCC — the strongly connected core is a small world (~16 clicks between random page pairs within SCC).
- **Short paths do not exist** from OUT back to SCC, or between tendrils.
- **Greedy search is harder** because you can only follow outgoing links — you cannot "pull" information from a page that doesn't link to you.

Search engines (Google, 1998) solve the navigability problem by building a *global index* — they observe the entire graph and compute optimal paths. This is the opposite of Kleinberg's decentralized search: it works precisely because it is *centralized*.

Takeaway

The Web's directed links create asymmetric reachability. Only the Strongly Connected Component (~28%) is a "small world." Search engines solve the navigability problem through centralized indexing — the opposite of Milgram-style decentralized routing.

Quick Check

A directed network has 1000 nodes. The SCC contains 300 nodes. The IN component has 200 nodes, and the OUT component has 250 nodes. The rest are tendrils.

1. From a random node in IN, can you reach a random node in OUT? Explain your path.
2. From a random node in OUT, can you reach a node in IN?
3. What fraction of all possible source-target pairs have a directed path between them?

Speaker notes

Give them 4 minutes. (1) Yes: IN → SCC → OUT. The IN component can reach the SCC by definition, and the SCC can reach OUT by definition. So IN → SCC → OUT is a valid directed path. (2) No — OUT cannot reach back to the SCC (by definition), and from the SCC you can reach OUT but not IN. OUT → IN has no path. (3) This is harder. Pairs within SCC: 300×299. IN→SCC: 200×300. IN→OUT via SCC: 200×250. SCC→OUT: 300×250. Others require more analysis. The point is that far fewer than 50% of all pairs are mutually reachable.

≡

Quick Check — Solution

IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are mutually reachable.

Quick Check — Solution

1. Yes: IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are mutually reachable.

Quick Check — Solution

1. Yes: IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

2. No. OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are mutually reachable.

Quick Check — Solution

1. Yes: IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

2. No. OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are *not* mutually reachable.

≡

Wrap-Up — The Five Gaps

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance
L07	Navigational	Short navigable paths	Local info only; directed links	Geometric link distribution; centralized search

The small-world phenomenon says short paths *exist*. Watts-Strogatz explains *why*. Kleinberg explains *when* they are *findable*. Milgram and Dodds et al. tested *whether* real people find them — and the answer is: sometimes, with effort, and only when the network's geometry cooperates.

Looking Ahead: Lecture 08

How do things *spread* on networks — diseases, information, behaviors? The structure we have built (ties, communities, balance, small-world shortcuts) now becomes the substrate for dynamic processes: contagion, diffusion, and cascading failure.

Reading

Chapter 20 of Easley & Kleinberg (2010) covers the small-world phenomenon and Watts-Strogatz. Kleinberg (2000), *The Small-World Phenomenon: An Algorithmic Perspective*, is the source for the decentralized search theorem. Dodds, Muhamad & Watts (2003) is the email experiment. Broder et al. (2000) is the bow-tie structure of the Web.

Speaker notes

The five-gap table closes the L03–L07 arc. Lecture 08 introduces the final conceptual shift: from structure to dynamics. The graph is no longer just an object to analyze — it becomes a *medium* through which processes propagate. The structural properties we have studied (weak ties, communities, balance, small-world shortcuts) now determine how fast and how far a contagion or cascade spreads.

Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/